Voice Controlled Wheelchair

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***Abstract*—The require for programmed Wheelchair is particularly show in care of undaunted individuals (individuals with tireless vegetative state, paraplegia, stroke and spinal line wounds), where the care requires a part of time and labor. This report is the result of a plan and advancement of an computerized multifunctional Wheelchair that would perform all capacities display in today's Wheelchair (Wheelchair with flexible parcel of back rest and leg rest additionally farther control with which we are able give all fundamental development) as well as modern capacities of fitting Wheelchair segments (leg positions altering). It is anticipated that this modern programmed Wheelchair would empower people's superior therapeutic care, and would significantly decrease time and labor to the old-age domestic staff. Hence, The purpose of this model is to plan, design and build a wheelc hair audio control system. The frame of the wheelchair include a motor frame and a voice recognition module controlled by a microcontroller. Hence, the created voice controlled wheelchair can give simple get to for individuals with physical inability additionally offer programmed security from impediment collision in case the botch of any voice commands happens**

Keywords— Speech recognition system, Wheelchair, Smart wheel chair, Automatic Wheel Chair

I. INTRODUCTION

A wheelchair is an important means of transportation for people who are physically unable to move anywhere. There are generally two types of wheelchairs. The main purpose would be a self-controlled chair, where the user must operate the chair with their hands and arms using the wheels on the outside of the wheelchair. The physical system may consist of a set of controls, such as joystick control. However, both wheelchairs require hands and arms. This makes it difficult for people without hands and arms to use a wheelchair. After that many researchers proposed specific ideas to control wheelchairs, but another possible idea is to use voice to control wheelchairs because the type of communication is more voice. Programmable wheelchairs have joysticks, remote controls, etc.

II. EASE OF USE

A. Speech Recognition System

Speech recognition is divided into two main groups: talker and talker. The speaker's relationship to the frame is prepared by the user who understands the frame. This method can achieve a high level of evaluation and confirm an accuracy of approximately 98%. They are the most used in most applications. Voice autonomy is the principle that allows anyone to respond with a single word, regardless of who is speaking. Therefore, the framework should respond to various aspects of speech production, expression, and speech purposes. [3] Technology often requires voice control to recognize speech. Speech recognition also includes speech patterns it can recognize. There are three types of speech: social speech, social speech and informal speech. The design presented in this article is based on the HM2007 control circuit (IC). The cards used are PLCC-48 square components. This could be a CMOS speaker, known as an LSI (Large Scale Integrated Circuit) circuit. The front-end features simulation, speech detection, control and power management. Supports external SRAM for storing programs and text. Time setting for 40 words and 0.96 seconds duration or 20 words and 1.92 seconds duration. External receiver accessories included for audio input. The response time of the chip is less than 300 milliseconds. A 3V battery is used to record preparation and test data after the circuit is turned off. Objective knowledge of speech depends on the speaker.

B. Automatic Wheelchair Systems

Wheelchairs can be considered one of the most important parts of technology.Their operating rules are similar to portable robots that can be used to explore various situations. There may be two wheels (two wheel adjustable) that are shown to need maintenance [3]

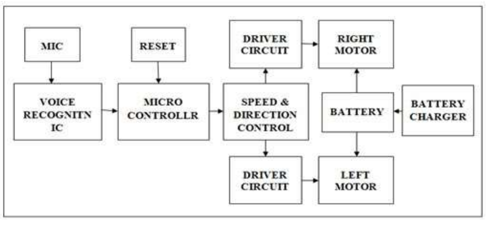


Fig.1. Block Diagram of voice controlled Wheel Chair

III. HARDWARE

A. Wheelchair

It has two 14A/24V/200W DC motors and two rollers for adjustment. Of course, he can also move the chair mechanically by pushing it with the help of his partner. If the engine starts, it accelerates to full throttle. If it is at this time, the engine will not stop easily.

B. Wheelchair Control Power Supply

The control power supply for the wheelchair motor, motor driver, control and other working times is two batteries, forming a 24V power supply, which can use 12V, as shown in the picture. The total conduction current on the seat is determined as 28A for both motors. Use the 7805 voltage controller to convert the 12VDC battery to 5VDC for use by the Arduino and motherboard.

C. Motor

As mentioned recently, there are two DC motors on the chair, details are as follows:

24VDC, 14A, 200W, 3900RPM, draw frame clutch, adapter as shown in the picture. The motor driver is an H-bridge circuit for two inductive loads to control the speed and altitude of each motor. The driver used is displayed here. It receives a PWM signal from the Arduino microcontroller, its duty cycle depends on the voice command and it also starts the motor. The brain of the wheelchair is a microcontroller that controls the movement of the chair, which is being considered as an option.

D. Amplifier

A good microphone is important when using programmed speech recognition (ASR). The microphone must be able to remove ambient noise that can complicate the ASR process. Fashionable headphones are the best choice with Hi-Fi innovation that reduces ambient noise.

E. Arduino

Arduino is an open source hardware and software community of companies, businesses, and consumers that design and manufacture microcontroller boards and microcontroller units for home computerized devices. The work is licensed under the CC BY-SA license and the program is licensed The development and distribution of Arduino boards is licensed under the GNU Lesser Public License (LGPL) or GNU Common Open License (GPL) [1], Author of the program: Kim. Arduino boards are sold through the official website or through authorized resellers.

Fig.2. Arduino Uno Processor

As we discussed, Arduino Uno is the most advanced board available and is the best choice for beginners. We can directly connect the development board to the computer with a USB cable, the USB cable works both as a power supply and serial port.

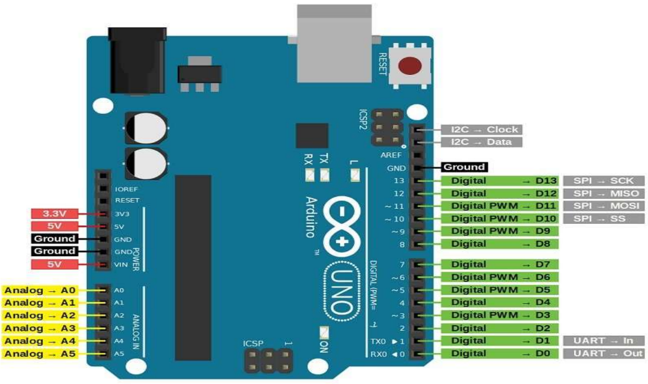


Fig.3. Arduino Uno

IV. DC MOTORS

DC motor is a motor that converts electrical energy into electricity. Most genres depend on the power provided by the area of interest. Almost every type of DC motor, whether electromechanical or electric, sometimes has some internal device that changes the flow of current through parts of the motor. [12]

DC motor is the widely used main motor as they need to get the fuel from the existing DC power source. Motors are widely available to work on existing integration but may be a light brush with multi-purpose controllers and systems. The simple DC motor consists of a series of permanent magnets in a stator and an armature with one or more protective metal windings around the elegant pressing center that concentrates the gravitational field. The windings usually have a different number of parts and on large motors there may be some current connections. The closed end of the winding is associated with the commutator. The commutator ensures that each armature coil is energized in turn and connects the pivot coils external control power via the brush.

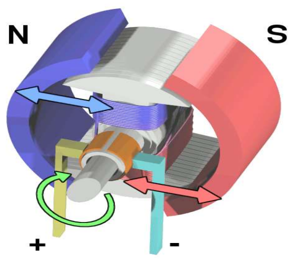


Fig.4.1 DC MOTOR PARTS

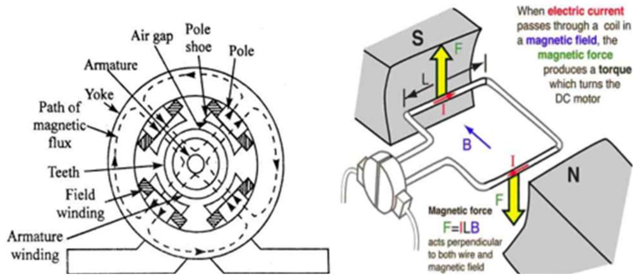


Fig.4.2 DC MOTOR PARTS

V. HC 05 Bluetooth Module

- The HC-05 module is designed for wireless communications and is one of the great addition to your model.
- This model can be used in master or slave mode depending on your specific needs.
- Used in many devices such as wireless headphones, game controllers, mice and keyboards.
- The range of this module can go up to <100m, although it does rely on various factors like the transmitter, environment, as well as geographical and urban conditions.
- Operating on the IEEE 802.15.1 standardized protocol, it enables the creation of wireless Personal Area Networks (PAN), facilitating the exchange of data using frequency-hopping spread spectrum (FHSS) radio technology.
- By leveraging serial communication, this module interfaces with devices through a serial port (USART).
- Bluetooth serial modules enable seamless communication between various serial-enabled devices via Bluetooth connectivity.
- It comes equipped with 6 pins:

1. Switch/EN: This pin plays an important role in converting the Bluetooth module to AT command mode. If set to high, the module runs in command mode. Otherwise it remains in file mode. The normal baud rate of the HC05 is 38400 bps in command mode and 9600 bps in data mode.

There are two different types of HC-05 module:

1. File format: Easily exchange data between connected devices.

2. Command mode: This mode uses AT commands to configure the HC-05. It relies on the module port (USART) to send these commands.

2. VCC: For best performance, be sure to connect this pin to 5V or 3.3V.

3. GND: This connection, which is the ground pin of the module, is very important for the correct operation of the module.

4. TXD: Responsible for wireless transfer of serial data (data received by the Bluetooth module is transmitted via the TXD pin).

5. RXD: Designed to receive data in text format (received data is transmitted wirelessly by the Bluetooth module). 6. Status: Provides an indication of the connection status of the given module.

HC-05 module manual

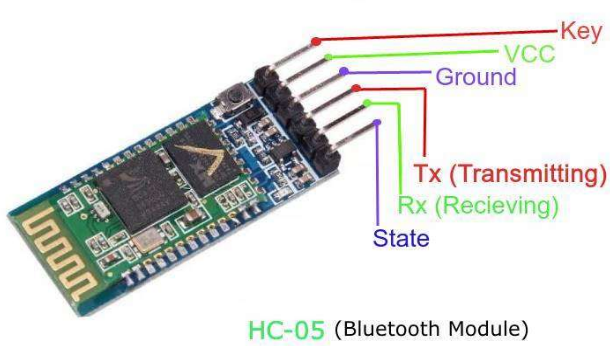


Fig.5. HC05 Bluetooth Module

Relay Module

Relays are often used when the circuit needs to be controlled by an autonomous low-power signal or when there are multiple circuits to be controlled by one signal. Historically, relays were used as signal repeaters on long-distance power lines; This involves amplifying the signal by sending it from one circuit to another. They were often used to perform useful tasks in mobile phones and early computers [15]

These versatile switches play an important role in the operation of various electronic devices and machines.

Fig. 6. Relay module

Methodology

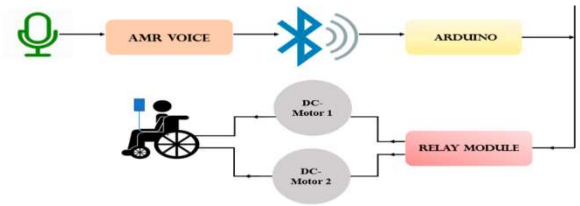


Fig. 7. flow diagram

Voice Recognition Module: Introduction

The language recognition module forms the basis of this model and is used to edit and display the desired language. It is divided into three levels: voice customization, voice detection and voice recognition. Audio customization is the process of combining audio recording needs with signal needs. Voice capture is the stage where the target's voice command is recorded and the voice is saved as a custom location. [6] The voice recognition stage is the last stage after the voice command is completed, the module will send a special signal to the microcontroller to perform the necessary actions. This figure shows the block diagram of the speech recognition module.

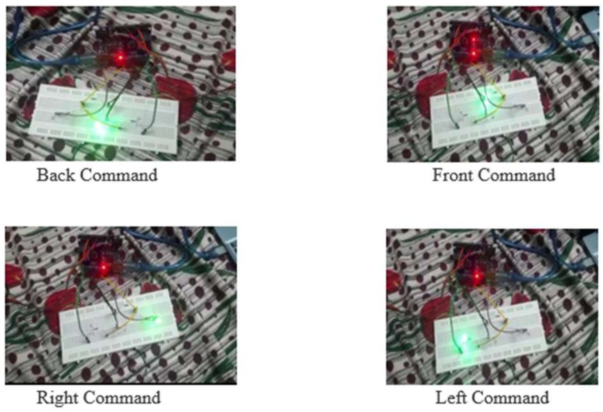
* Voice customization level
* Voice capture level
* Voice recognition level

Using serial communications computer to write voice commands Access Point connection language software, baud rate up to 9600 . Then, if the voice recognition module is connected to the computer, the current voice command is canceled by sending the hexadecimal command AA 01. Start recording by sending the voice message in Group 1 (hex AA 11 command). After sending this command, the user must type the five-letter number to end the record group. Once the voice is recorded, verification must include group 1 in voice recognition by sending the hexadecimal AA 21 command. Before recording, double-check the recording by repeating these five commands. Figure 3 shows that the module checks whether all commands are valid and then emits a confirmation tone. The five voice commands are: forward, backward, turn left, turn right, and stop. HC-05 Bluetooth module works as wireless communication between microcontroller and voice recognition module to control the power of the wheelchair. This will eliminate the need for long, messy pipes.

The document describes the success of the electric chair operated by joystick or voice recognition. Voice recognition works on most orders (over 95%). Designing a simple and effective automatic speech recognition system to control the electric chair for disabled people is the interest of this article. Rather than storing the entire speech signal, the data generator stores image vectors to match the speech processing time; This saves storage space and makes the pairing process very fast. The operating system (audio equipment and microcontroller) is directly connected to the wheelchair in a single package, leading to the realization of intelligent wheelchair design. The speaker was tested to check its performance in creating the movement of the chair. It has been confirmed that the acceptance rate has reached 95%. The system will only recognize a word if it is not pronounced correctly. In general, users are satisfied with the system. The Arduino-based voice command wheelchair has been carefully designed and tested to respond to commands. The quality of life of disabled people will increase. Costs are also kept low by attaching the design to each wheelchair.

The model provides the following findings:

1. Speech recognition module is working.
2. The DC motor must work and the motor driver must work.
3. Connect the speech recognition module to the microcontroller.
4. Relay works properly



. Fig.8. Commands

Conclusion

In conclusion, the *development* of a voice-automated wheelchair represents a significant advancement in accessibility and mobility solutions for individuals with mobility impairments. By integrating voice recognition technology with wheelchair control systems, this innovative project offers users a seamless and intuitive method of *navigation* and control. Through *meticulous research*, design, and prototyping, we have addressed the unique needs and challenges of wheelchair users, prioritizing safety, reliability, and ease of *use*, As we continue to refine and iterate upon our design, we envision a *future* where voice-automated wheelchairs empower individuals to navigate *their* environments with greater independence and dignity. The initiative highlights the potential of technology to improve the quality of life of people with disabilities and the importance of design in creating solutions for everyone.

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